

LIFENG CHEN

✉ lifengchen299@gmail.com · ☎ +1 706-202-7889 · in CCConstant · 🌐 clf28.github.io

🎓 EDUCATION

University of Georgia

Ph.D. in Computer Science

Aug. 2026 – Present

Beijing Jiaotong University

B.Sc. in Computer Science

Sep. 2022 – Jun. 2026

GPA: 3.9/4.0

📄 PAPERS

1. **Detail++: Training-Free Detail Enhancer for Text-to-Image Diffusion Models.**
Lifeng Chen, Jiner Wang, Zihao Pan, et al.
Project page **IEEE Transactions on Image Processing** 2025
2. **ECHO: Efficient Chest X-ray Report Generation with One-step Block Diffusion.**
*Lifeng Chen**, Tianqi You, Hao Liu, et al.
Project page **ECCV 2026** 2026
3. **Discrete Diffusion Models with MLLMs for Unified Medical Multimodal Generation.**
Jiawei Mao, Yuhan Wang, Lifeng Chen, et al.
Project page **ECCV 2026** 2026
4. **Genesis: Visual Augmentation Primitives for Multimodal Chain-of-Thought Reasoning.**
*Fan Yang**, *Lifeng Chen**, Yipu Wang, et al.
EMNLP 2026 Main Conference 2026

*Equal contribution.

🔬 RESEARCH EXPERIENCE

Xiaohongshu, Super Intelligence Team

Research Intern, Multimodal Foundation Models

Mar. 2026 – Aug. 2026

Advisor: Feilong Chen

- Developed **Genesis**, using GPT-Image-2 to generate answer-changing STEM training pairs through visual-primitive editing, with primitive-grounded reasoning traces. Trained MLLMs from **4B to 32B** using rubric-based multimodal CoT RL with primitive-level rewards along these traces; the 4B model outperformed 8B baselines, while the training pipeline raised an internal MLLM's ZEROBench_{sub} score from **23.12 to 32.33**.
- Proposed the **UNREAL Benchmark**, a multi-subject multimodal STEM benchmark that tests reasoning from visual evidence when scenes contradict commonsense and textbook priors.
- Co-designed **CoAnchor**, injecting priors from a frozen geometry foundation model into cross-view attention and progressively training MLLMs to use point correspondences as explicit CoT evidence; outperformed GPT-5.2 by **9.8 points** across four multi-view benchmarks.
- Evaluated the dots-based model for **WenYiWen**, Xiaohongshu's AI search feature that answers user questions by searching and summarizing community posts. Assessed whether it found relevant posts and accurately summarized their content.

Midea Group, AI Research Center

Research Intern, Medical Multimodal Foundation Models

Aug. 2025 – Mar. 2026

Advisor: Hao Liu

- Led **ECHO**, converting a pretrained autoregressive MLLM into a block-diffusion model using only **2.2%** of its SFT corpus; reduced adaptation FLOPs by **72.3%** and achieved **5.1×** practical speedup with one-step-per-block decoding. To retain generation quality, developed trajectory-level distillation that combines token distributions from the teacher's multi-step denoising trajectory into a joint target, preserving cross-token dependencies otherwise lost under mean-field prediction.
- Engineered full-parameter post-training of a **32B MLLM** with Megatron-LM across data, tensor, pipeline, and sequence parallelism, with precision alignment. Evaluated mixtures of general-purpose and medical training data and varied positive-to-negative clinical case ratios.
- Built a multimodal agent with image-manipulation and DICOM tools, and trained its tool-use policy with reinforcement learning for medical image reasoning.
- Built a Chinese-language semantic reward with LLM-based NER and Qwen embeddings; added dynamic sigmoid calibration to prevent vanishing gradients and selected 16 high-signal trajectories from 128 rollouts for advantage estimation.

University of California, Santa Cruz, VLAA Lab

Mar. 2025 – Jul. 2025

Research Intern, Unified Multimodal Modeling

Advisor: Yuyin Zhou

- Co-designed **MeDiM**, a unified 7B discrete-diffusion MLLM that generates medical images, reports, or both with one shared backbone; adapted a pretrained MLLM and owned DeepSpeed training, improving downstream report generation by up to **31.6%**.
- Built an automated pipeline for seven visual conditions, including depth, segmentation, and edges; fused MM-DiT/U-Net features with CLIP semantics to strengthen MLLM spatial understanding.

Westlake University, AGI Lab

Oct. 2024 – Feb. 2025

Visiting Student, Controllable Generation

Advisor: Chi Zhang

- Proposed and led **Detail++**, a training-free framework that decomposes complex prompts into subject-specific sub-prompts and progressively injects attributes into their corresponding latent regions, reducing attribute leakage and style blending; improved attribute-binding performance by **15.9%** on SDXL, **11.1%** on SD3, and **16.3%** on Flux.
- Developed continual-learning personalization with concept-specific LoRAs and orthogonality regularization to reduce interference when composing independently learned concepts.

⚙️ SKILLS

-
- **Frameworks and systems:** PyTorch, Megatron-LM, DeepSpeed, Linux, distributed foundation-model training
 - **Research areas:** LLM/MLLM foundation models, agentic RL, multimodal reasoning, unified multimodal models, discrete diffusion language models, spatial intelligence